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	Test plan
PROJECT NAME: 702 – Skincare	
FILE NAME: 702-pumpcontrol102 Test plan	RELEASE

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1. Scope

The purpose of this document is to settle a step-by-step plan for testing the device's functionality.

2. References

3. Equipment

- Oscilloscope – Tektronix MDO3024;
- Multimeter – Fluke 87V;
- IR imager – Agilent U5855A;
- DC power supply – Keysight U8031A.

4. Test conditions

The test conditions are valid for all tests unless otherwise specified.

- Ambient temperature - $23^{\circ}\text{C} \pm 3^{\circ}\text{C}$
- Relative humidity - 30÷60 %
- Atmospheric pressure - 700÷1060 hPa

5. Visual inspection

Perform visual inspection and check for anything unusual on the PCB. Do NOT connect the PCB to the Power supply during this test.

Table 1. Visual inspection check list.

Component checked	What to look for	Result
Labels	Missing/Incorrect/Incomplete	Pass/Fail
Solder joints	Missing/Cold solder	Pass/Fail
Components	Missing/Damaged/Misaligned	Pass/Fail
Short circuit	PCB damage/Bad soldering	Pass/Fail
PCB solder mask	Missing/Damaged	Pass/Fail

6. Electrical measurements

HIGH signal level is defined as a voltage of $>2.5\text{V}$.

LOW signal level is defined as a voltage of $<0.5\text{V}$.

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6.1. Power

Stage 1: Pump/Fan OFF

- Connect a +24V power supply to J1.
- Confirm +24V_FILT is within limits [22V...26V] (TP3/TP6).
- Confirm +15V is within limits [14.19V...15.70V] (TP19).
- Confirm +3.3V is within limits [3.16V...3.48V] (TP11).
- Confirm UVEN is LOW (TP15).
- Confirm +24V is disabled ($\approx 0V$) (TP1).
- Confirm DL1 is OFF and +24V_FAN is disabled ($\approx 0V$) (TP4).
- Confirm \PG is HIGH (TP16).

Stage 2: Pump/Fan ON

- Disconnect the +24V power supply connector (J1).
- Tie ON/O\F\F\ (TP12, J5/4) to HIGH.
- Connect a fan to J2.
- Connect a pump to J3.
- Measure the SOA of T1: Reconnect the +24V power supply connector (J1) and, via R1, measure the current transient through T1 using the oscilloscope. Save the graph and include it in the report.
- Confirm UVEN is HIGH (TP15).
- Confirm +24V is within limits [22V...26V] (TP1).
- Confirm DL1 is ON and +24V_FAN is within limits [22V...26V] (TP4).
- Confirm \PG is LOW (TP16).
- Disconnect the +24V power supply connector (J1).
- Remove the ON/O\F\F\ (TP12, J5/4) connection.

6.2. Driver

Stage 1: Pump/Fan OFF

- Connect a programming connector to J4.
- Upload the firmware to the driver.
- Confirm DL2 is ON.
- Disconnect the programming connector.
- Reconnect the +24V power supply connector (J1).
- Confirm BUS_SENSE is LOW (TP17).
- Confirm VSP is LOW (TP22).

Stage 2: Pump/Fan ON

- Disconnect the +24V power supply connector (J1).
- Tie ON/O\F\F\ (TP12, J5/4) to HIGH.
- Reconnect the +24V power supply connector (J1).

- Confirm BUS_SENSE is within limits [1.96V...2.41V] (TP17).
- Confirm VSP is within limits [2.70V...2.84V] (TP22).
- Using the oscilloscope, confirm the PWM frequency at SPEED_FB is within limits [330Hz...350Hz] (TP10). Save the graph and include it in the report.
- Using the oscilloscope, capture the voltage waveforms at U (TP9), V (TP8), and W (TP7) and include them in the report.
- Using the oscilloscope and current clamp, capture the current waveforms at U (J3/1), V (J3/2), and W (J3/3) and include them in the report.
- Disconnect the +24V power supply connector (J1).
- Remove the ON/O\F\F\ (TP12, J5/4) connection.

6.3. Control

Stage 1: Pump/Fan OFF

- Reconnect the +24V power supply connector (J1).
- Momentarily connect \FAULT_RST to LOW and then disconnect it (TP13).
- Confirm \FAULT_RST is HIGH (TP13).
- Confirm ON/O\F\F\ is LOW (TP12).
- Confirm \FAULT is HIGH (TP14).

Stage 2: Pump/Fan ON

- Disconnect the +24V power supply connector (J1).
- Tie ON/O\F\F\ (TP12, J5/4) to HIGH.
- Reconnect the +24V power supply connector (J1).
- Momentarily connect \FAULT_RST to LOW and then disconnect it (TP13).
- Confirm \FAULT_RST is HIGH (TP13).
- Confirm ON/O\F\F\ is HIGH (TP12).
- Confirm \FAULT is HIGH (TP14).

7. Temperature measurements

Using the IR imager, measure the temperature of the board when it reaches temperature equilibrium. Take pictures and include them in the report.

8. Conclusion

Compare the results from all measurements. Based on the results and the calculations in the design report, make a conclusion.